

HW-1: Limit counter

A counter that stops itself

Count up while EN is high and clear the count at LIMIT.

Keep the module and port names of the skeleton exactly.

Interface signals

CLK : Core clock

nRST : Active-low reset, asynchronous

EN : Count on this edge while high

LIMIT : The count the run ends at

CNT : The count now held

DONE : High for the cycle the count reaches LIMIT.

Detailed operations

Hold nRST low to clear the counter at once.

The counter advances only on a rising edge with EN high.

The count returns to zero on the edge after it reaches LIMIT.

Leave the given folder as it is: one flat directory.